

Lê Tiến Phát

COMPUTER ENGINEERING | EMBEDDED SYSTEMS & IoT

Ho Chi Minh City, Vietnam | letienphatwork@gmail.com
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Computer Engineering studies completed at HCMUT; diploma pending. Hands-on internship experience in IoT device integration, firmware troubleshooting and monitoring systems, complemented by academic work in anomaly detection and edge deployment. Seeking entry-level embedded, testing, manufacturing technology or technical implementation roles.

PROFESSIONAL EXPERIENCE

IoT Development Intern | Lotus Vietnam

Lotus Viet Nam Electronic Service - Trading - Manufacturing Co., Ltd. | Jun - Aug 2025

- Integrated ESP32-S3 controllers and environmental sensors into a monitoring pipeline using MQTT, Node-RED and Grafana.
- Developed custom Tasmota drivers for RS485/Modbus devices; troubleshoot communication, configuration and hardware-integration faults.
- Tested continuous operation and sensor-data stability; documented device configurations and troubleshooting procedures for repeatable deployment.

ACADEMIC & INDEPENDENT PROJECTS

SCADA / ICS Anomaly Detection with Edge Deployment

Academic capstone | HCMUT | 2026

- Built a configurable signal simulator with replay and fault injection to support repeatable anomaly-detection experiments.
- Deployed an anomaly-detection model with TensorFlow Lite on Raspberry Pi and evaluated an Isolation Forest on ESP32-S3 using emlearn.
- Investigated preprocessing and false-alert issues; compared detection performance, inference latency and hardware constraints, and documented findings.

DeskFlow | Workspace Booking and Pricing Platform

Independent project | 2026 | github.com/Ixalis/Deskflow

- Built a FastAPI/SQLAlchemy backend and TypeScript client; implemented automated tests and investigated booking, timezone and revenue-rounding defects.

TECHNICAL SKILLS

Programming: C/C++, Python, SQL

Embedded & IoT: ESP32-S3, Raspberry Pi, Tasmota, RS485/Modbus, MQTT

Tools & testing: Git, Linux, Node-RED, Grafana, Pytest, fault injection, troubleshooting

Analysis & communication: Signal processing, model evaluation, technical documentation, English communication

EDUCATION

Computer Engineering | Ho Chi Minh City University of Technology

HCMUT, VNU-HCM | 2022 - 2026 | Studies completed; diploma pending.